

No. of Functional Features Included in the Solution

Date	15 July 2024
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Interface	Allows students to enter learning field and problem statement	Streamlit UI	Done	High
2	Emotion Detection	Predicts emotions using BERT and BiLSTM models	Emotion Classifier	Done	High
3	Mixed Emotion Analysis	Detects primary and secondary emotions with confidence scores	Emotion Analysis	Done	High
4	AI Learning Guidance	Generates personalized learning suggestions using Gemini API	Gemini Service	Done	High
5	Emotion History Storage	Stores prediction history and student records	SQLite Database	Done	Medium
6	Analytics Dashboard	Displays emotion statistics, charts, and trends	Plotly Dashboard	Done	High
7	CSV Logging	Maintains backup of prediction records	CSV Storage	Done	Medium
8	Session Management	Maintains user interaction during application usage	Authentication / Session Manager	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Emotion Detection, AI Guidance, Mixed Emotion Analysis, Analytics Dashboard

Additional / Nice-to-Have Features	CSV Backup, Session Management, Confidence Score Visualization
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Student Input Form, Analytics Dashboard
2	Backend / Logic	3	Emotion Detection, Mixed Emotion Analysis, Confidence Calculation
3	Database / Storage	2	SQLite Storage, CSV Logging
4	API / Integration	1	Google Gemini API Integration
5	Security / Authentication	1	Session Management